

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 1: Brainstorming & Ideation — User Empathy Maps

Project Name: Cab Booking (`UCab`)

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1. Overview of Empathy Mapping

To design a user-centric cab booking platform, empathy maps were constructed for two primary stakeholders: the Urban Commuter (Sarah) who books rides daily, and the Fleet Dispatch Manager (Rajesh) who oversees cab operations and trip dispatches.

2. Persona 1: Urban Commuter (Sarah, 29 - Marketing Manager)

```
mindmap
  root((Sarah: Urban Commuter))
  Says & Thinks
    :: I need a reliable cab right now to catch my flight.
    :: Why are fare estimates so confusing and hidden?
    :: Can I get a clean sedan with drinking water?
  Does
    :: Compares cab prices across multiple apps on her phone.
    :: Checks driver ratings and vehicle number plates carefully.
    :: Downloads tax invoices for corporate expense reimbursement.
  Sees
    :: Cluttered interfaces with unexpected price surges.
    :: Long arrival delays without live driver updates.
  Feels
    :: Frustrated by hidden costs at checkout.
    :: Relieved when exact driver and car details appear instantly.
```

Empathy Breakdown Table — Sarah (Rider)

Quadrant	Key Observations & Insights	How UCab Addresses This
SAYS	"I want to know the exact fare breakdown before confirming my booking."	Transparent Pricing: Displays base fare (`150`), per-KM rate (`12/km`), and live promo discounts (`UCAB20`).
THINKS	"Is my cab actually coming, and who is driving?"	Live Dispatch Tracker: Shows real-time progress bar with assigned driver credentials (`Vikram Sharma - 4.9 & `).
DOES	Frequently travels with luggage after office hours and prefers extra comfort.	In-Ride Customization: Allows pre-ordering chilled mineral water and snacks (`+ 150`).
FEELS	Stressed about getting receipts for company tax claims.	Instant Corporate Receipts: One-click PDF thermal invoice download with QR verification code.

3. Persona 2: Fleet Dispatch Manager (Rajesh, 45 - Operations Lead)

```
mindmap
  root((Rajesh: Dispatch Lead))
  Says & Thinks
    :: I need to know our active fleet inventory instantly.
    :: How much total revenue have our cabs generated today?
    :: We need a simple tool to update ride status for drivers.
  Does
    :: Logs into admin portal first thing every morning.
    :: Toggles vehicles between Available and Busy status.
    :: Monitors incoming user bookings and dispatches drivers.
  Sees
    :: Fragmented Excel sheets and manual driver logs.
    :: Lack of centralized analytics on cab category performance.
  Feels
    :: Overwhelmed during peak morning booking hours.
```

:: Empowered when using a unified real-time executive dashboard.

Empathy Breakdown Table — Rajesh (Admin)

Quadrant Key Observations & Insights How UCab Addresses This

****SAYS**** "We need one clean screen showing revenue, total rides, and active cars." ****Analytics KPI Bar:**** Real-time top-level summary cards displaying Gross Revenue (¹), Total Bookings, and Active Users.

****THINKS**** "Can I quickly add a new vehicle when our fleet expands?" ****Complete Fleet CRUD:**** Intuitive `AddCar.jsx` and `EditCar.jsx` forms with image upload and instant status toggling.

****DOES**** Manages dispatch by updating ride states from pending to started. ****Trip Control Monitor:**** Centralized `AdminBookings.jsx` where any ride status can be updated with 1 click.

****FEELS**** Worried about database server crashes bringing down dispatch operations. ****Zero-Config Storage Engine:**** Automatically falls back to `data.json` if database services disconnect, ensuring zero downtime.